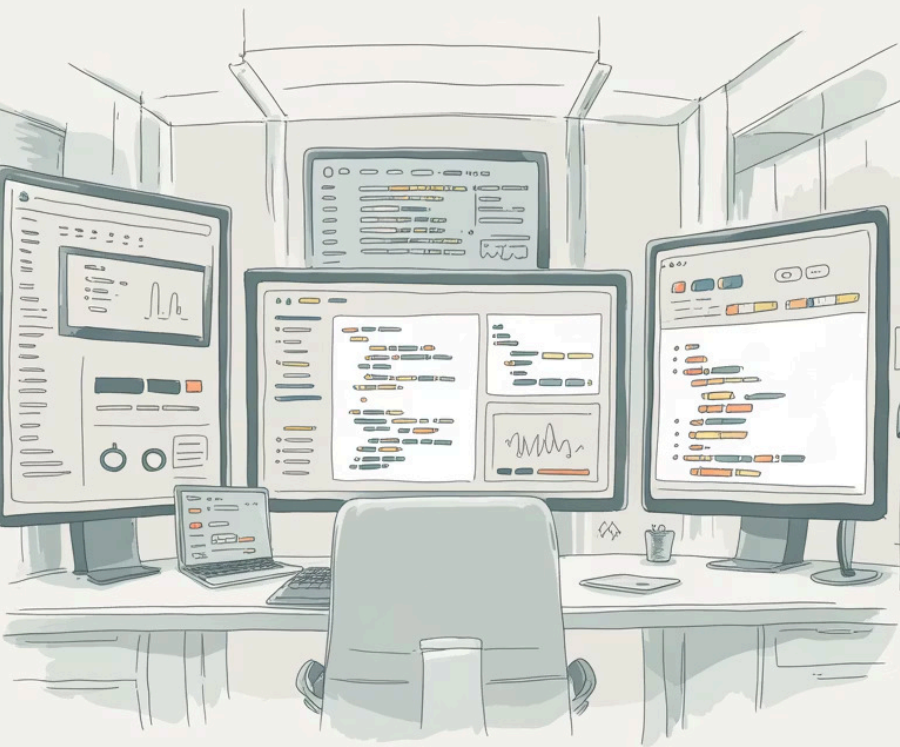




Build Faster

Automated pipelines and intuitive tools to
accelerate your development workflow

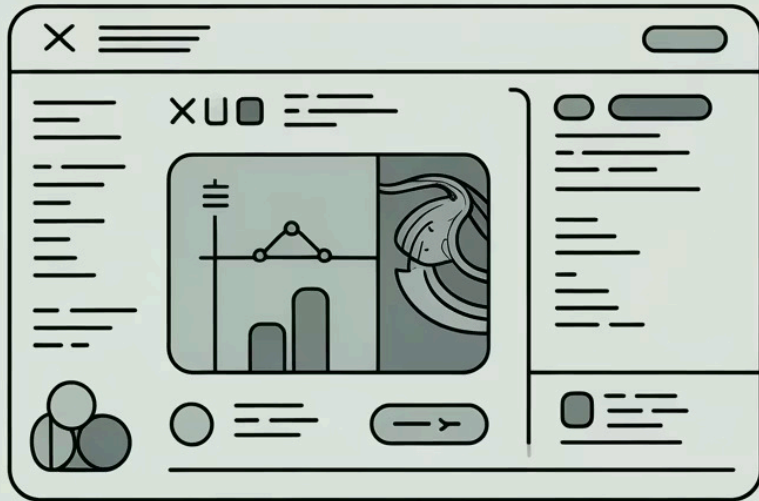
START FREE TRIAL

Introduction to Jenkins

Continuous Integration & Continuous Delivery (CI/CD)

Naresh Chaurasia

Jenkins

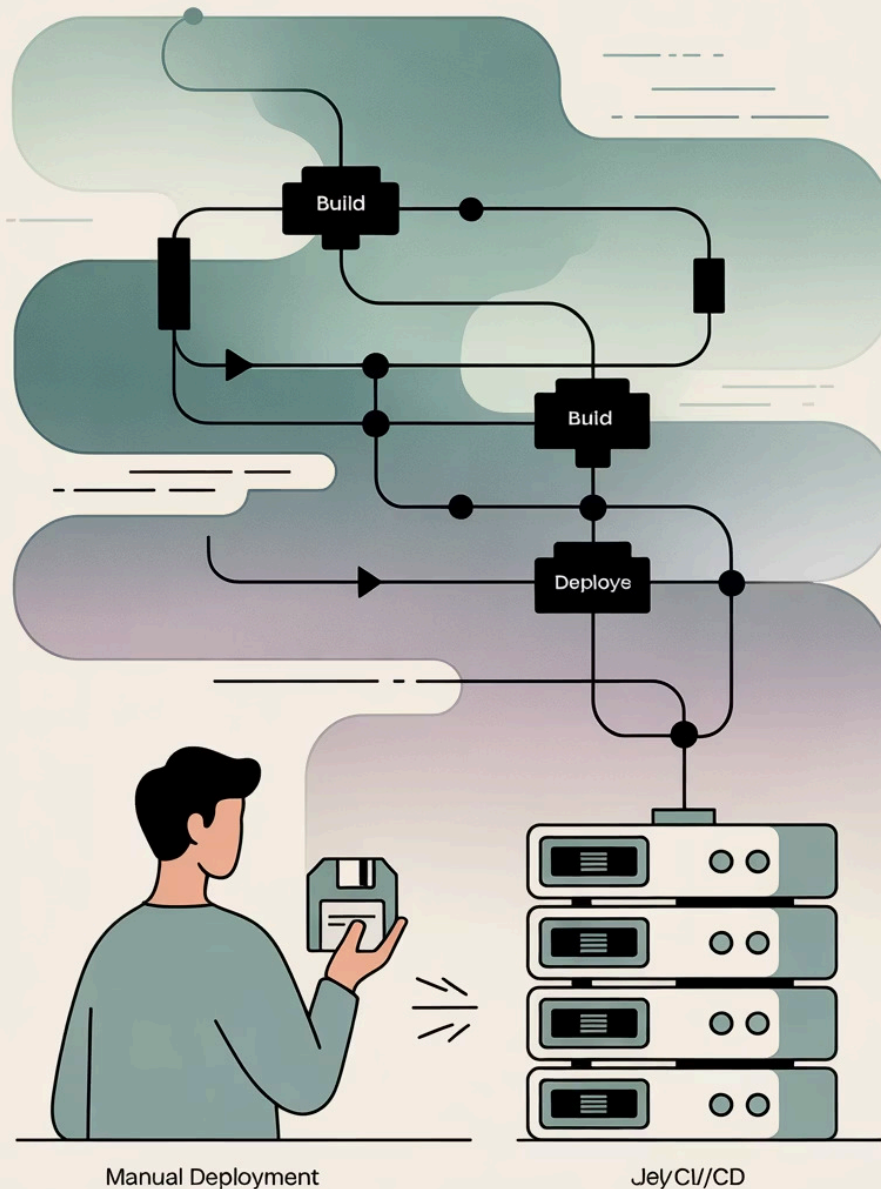


What is Jenkins?

Open-source automation server

Helps automate building, testing, and deploying applications

Widely used in DevOps & CI/CD pipelines



Why Jenkins? (Need for CI/CD)

The Problem

Manual builds and deployments are slow & error-prone

Developers need faster feedback on code changes

The Solution

Jenkins enables continuous integration and delivery

Jenkins Features



Easy setup and web-based UI



Works with Git, Docker, Kubernetes, etc.

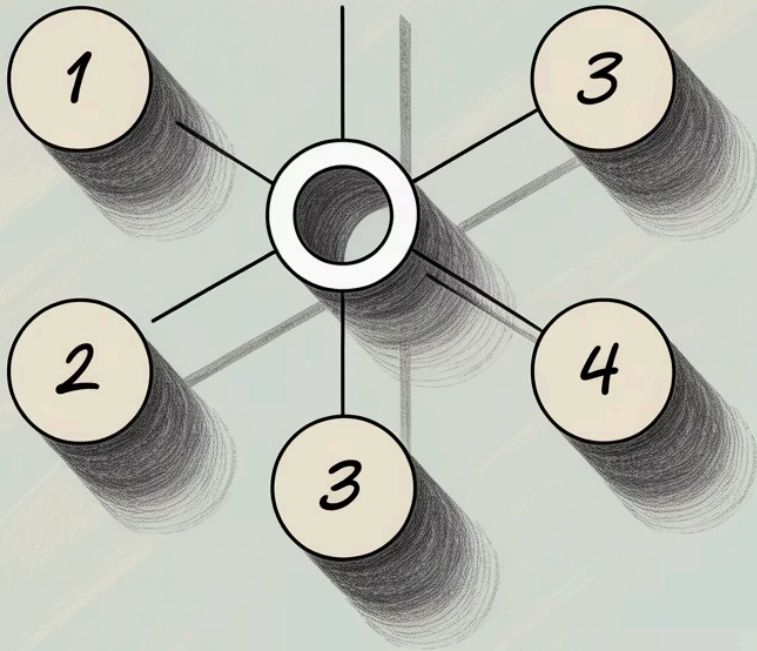


Supports 1000+ plugins

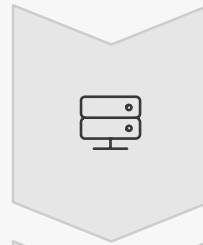


Scalable with master-slave architecture

Jenkins®



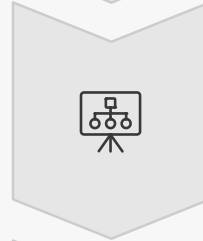
Jenkins Architecture



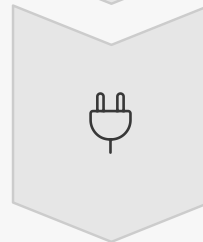
Master (controller)



Agents (workers/slaves)



Jobs & pipelines



Plugins for integration

Key Jenkins Concepts



Job/Project

Unit of work (build/test)



Pipeline

Sequence of steps for automation



Build Trigger

When job runs (push, schedule, etc.)



Workspace

Where job runs

Jenkins Pipeline (Code as Pipeline)

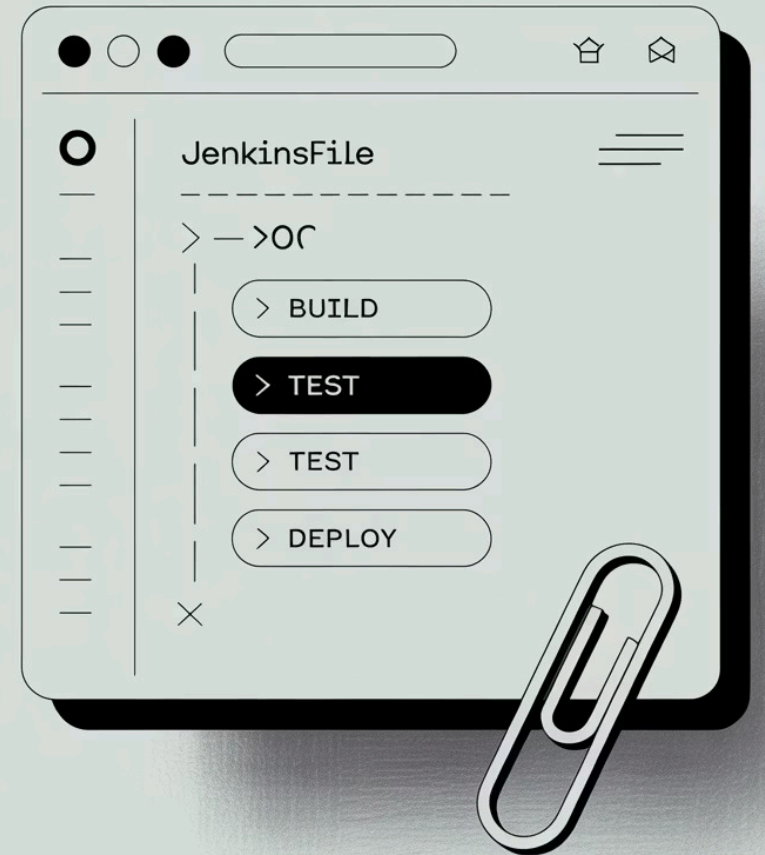
Pipeline Types

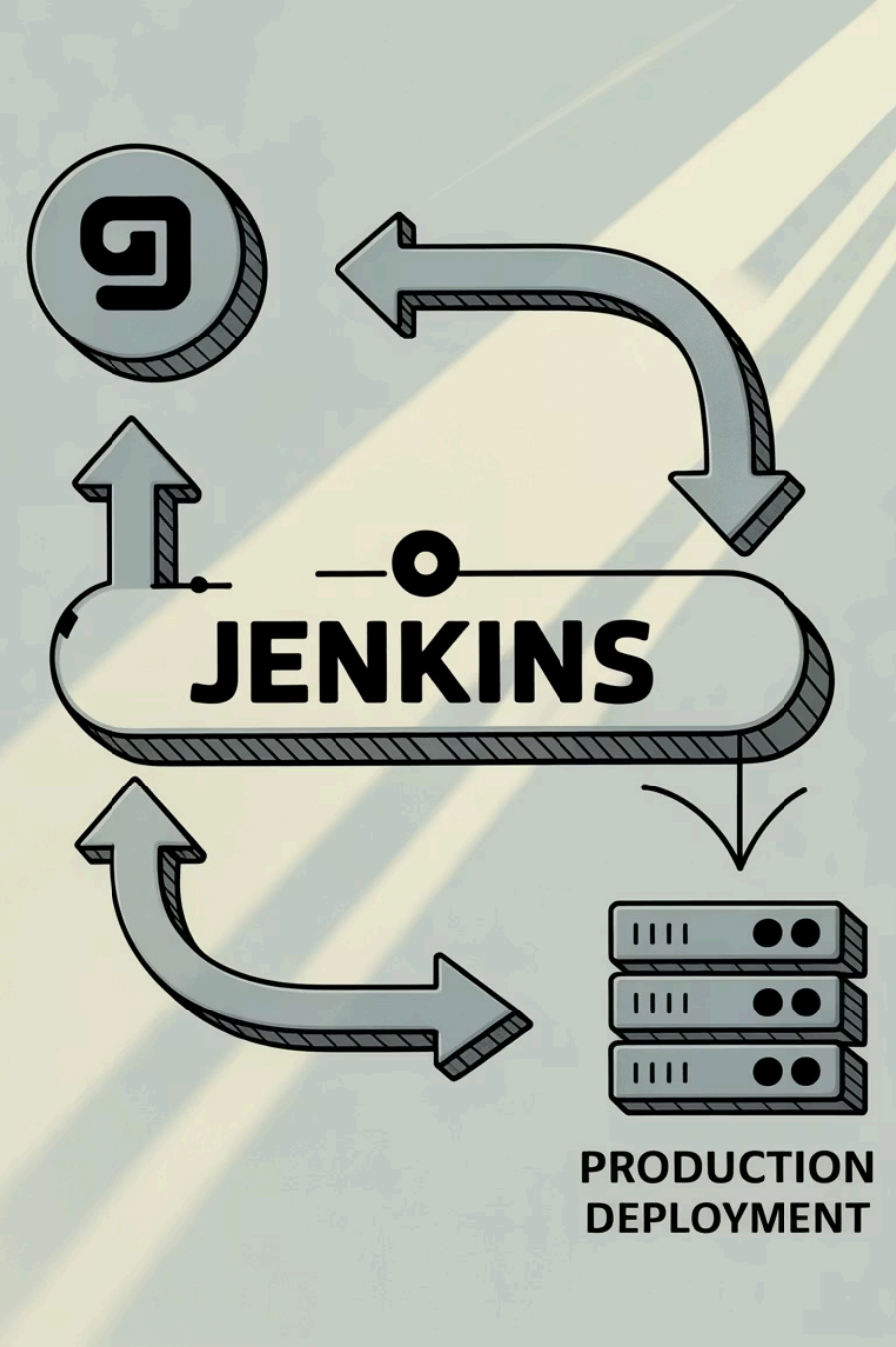
Declarative vs Scripted
pipeline

Jenkinsfile example

Benefits

- version control
- repeatability





Jenkins in Action (Demo Flow)

01

Code pushed to GitHub

02

Jenkins triggers build automatically

03

Tests run, artifacts created

04

Deployment to staging/production

Advantages & Use Cases



Faster development cycles



Early bug detection



Automated testing & deployment



Widely used in IT companies for DevOps

Conclusion & Next Steps

Jenkins is essential for CI/CD in modern development

Next Steps:

- Hands-on practice: setup Jenkins locally or on cloud
- Explore plugins and integrate with GitHub/Docker

